

Syed M Ali Naqvi

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Education

Habib University

BS in Computer Science

August 2021 – May 2025

- CGPA: 4.0/4.0
- **Minors:** Electrical & Computer Engineering, Mathematics
- President's Medal 2025, Dean's Medal (Computer Science)
- 100% 4-year Merit Scholarship

Stanford University

International Honors Program

June 2023 – August 2023

- GPA: 3.887/4.3
- **Coursework:** Machine Learning (CS229), High Performance Computing, Technological Entrepreneurship

Experience

AI and ML Engineer (R&D)

Farmevo Technologies

Karachi, Pakistan

April 2026 - Present

- Leading the design and development of the software stack for a **quadruped-based autonomous sensing platform**, integrating real-time navigation, localization, multimodal sensing, perception, and data recording for tree analytics, and forest inventory workflows.
- Developing a **LiDAR-camera 3D reconstruction pipeline** using point-based registration and LiDAR odometry/SLAM to generate accurate 3D representations of forest environments.
- Benchmarked **3D tree segmentation and parameter-estimation tools**, evaluating SOTA models including **ForestFormer3D**, **SegmentAnyTree**, **TreeLearn**, and **PointNet++** on curated forestry datasets to identify suitable models for automated tree-level analytics. Developing the pipeline for precise carbon stock estimation and tree analytics.
- Implemented **camera-LiDAR fusion** for point-cloud colorization and 2D-3D mask association, enabling integration of visual and geometric information for tree segmentation and downstream parameter estimation.
- Contributed to **mechanical and electrical integration of the quadruped forestry sensing platform**, including sensor selection, CAD design, 3D printing, and assembly of the onboard computer and multimodal sensor stack.
- Supervising an FYP research project on **2D tree segmentation and bark-texture-based species detection**, guiding dataset development, model selection, training, and evaluation for automated tree identification.

Autonomous Vehicle Algorithms Engineer

Covolv.ai

Karachi, Pakistan

June 2025 - March 2026

- Architected and optimized a **real-time multi-camera BEV perception pipeline in ROS 2**, fusing 6 synchronized camera streams via composable nodes and multithreaded execution to process high-resolution inputs at **25 FPS on embedded platform**.
- Fine-tuned vision-based deep learning models for **multi-object detection in L2+ autonomous driving**, spanning highway driving and autonomous parking; developed an **oriented bounding-box CNN** for parking-slot detection and owned the pipeline from data curation through training and evaluation.
- Engineered a inference pipeline for vehicle AI workloads, offloading neural-network inference from GPU to the **Hexagon Tensor Processor** and achieving an **87% improvement in end-to-end inference performance**.
- Optimized CNN perception models for HTP deployment through **post-training quantization**, systematically evaluating precision, dynamic ranges, and backend mappings; developed model-analysis tooling to quantify **layer-wise sensitivity to quantization** and guide mixed-precision deployment decisions.
- Designed **camera intrinsic and camera-to-camera extrinsic calibration workflows** using MATLAB Calibration Toolbox, establishing data-acquisition and calibration procedures for accurate multi-view geometric alignment and BEV projection.
- Developed a **Differentiable ISP-based domain adaptation pipeline** to improve perception robustness across heterogeneous camera sensors and low-light conditions, recovering **80-90% of performance lost under photometric degradation** while preserving baseline model behavior; owned dataset curation, training, fine-tuning, evaluation, and performance analysis.

Teaching Assistant

Habib University

Karachi, Pakistan

August 2022 - April 2025

- Selected as a **Teaching Assistant** for 6 courses across CS, mathematics, and AI, including **Deep Learning, Algorithm Design & Analysis, Computer Architecture, Linear Algebra, Discrete Mathematics, and Programming Fundamentals** during 2022-2025.

- Conducted **labs, recitations, and office hours** and designed, evaluated, and graded coursework, mentoring students in programming, algorithms, computer systems, mathematical foundations, and deep learning.
- Awarded **Best Teaching Assistant (Spring 2024)** by Habib University's Office of Career Services for outstanding teaching and student mentorship.

Undergraduate Research Assistant

Habib University

Karachi, Pakistan

September 2024 - April 2025

- Built an end-to-end computational pipeline for **CRISPR-Cas13b guide RNA design against all four Dengue virus serotypes**, processing and quality-filtering **38,057 viral genomes** and extracting **3.78M unique 30-nt candidate spacers** for conservation analysis.
- Designed a memory-efficient sequence-analysis workflow and **Cas13b-specific optimization algorithm** that tracked spacer conservation across viral populations and introduced up to 3 strategic nucleotide substitutions; improved the potency score of **98.5% of 1,740 candidates** by **47.17 points on average**.
- Developed and benchmarked ML models for **Cas13b targeting-efficiency prediction** using **245 experimentally validated spacers** and **157 engineered sequence features**, comparing XGBoost and other classical models against **Nucleotide Transformer** and **RNA-FM foundation-model embeddings**.
- Built a multi-objective ranking framework combining **predicted efficiency, serotype-specific conservation, and biochemical sequence constraints**, producing 269–396 high-confidence crRNA candidates per serotype suitable for experimental validation.

Summer Research Intern

Texas A&M University

College Station, Texas

May 2024 - July 2024

- Implemented and benchmarked **state-of-the-art spectral calibration method** for Swept-Source Optical Coherence Tomography (SS-OCT), establishing baselines for comparative evaluation.
- Developed a **deep learning-based spectral calibration pipeline using LSTM networks**, improving robustness to jitter noise and downsampling and reducing calibration error to **12%**.

Open Source Contributions

Autoware - Autonomous Highway Pilot

Autoware Foundation

- Contributed to developing **distance and speed estimation pipeline** for safety trigger module for an open-source highway autonomy system that powers safe autonomous driving on Highways.

Conference Papers

Computational Discovery of CRISPR Cas13b Guide RNAs for Broad-Spectrum Dengue Virus Targeting

March 2026

Presented at BICOB 2026 held in Hawaii, USA

S.M Ali Farhan Khan, Muslim Hussain

https://link.springer.com/chapter/10.1007/978-3-032-26028-4_18

Impedance-Based Crack Detection in Chemical Tanks

December 2024

Presented at 26th International Multi-Topic Conference (INMIC), Karachi, Pakistan

S.M Hussain, T. Mumtaz, **S.M Ali**

<https://ieeexplore.ieee.org/document/11004358>

Fine-tuned LLM based approach to Scientific Text Simplification

September 2024

Presented at CLEF 2024 Lab Proceedings held in Grenoble, France

S.M Ali H. Sajid, O. Aijaz, O. Waheed, F. Alvi, A. Samad

<https://ceur-ws.org/Vol-3740/paper-309.pdf>

Projects

KisanEye - Aerial-based Crop Health Monitoring System

- Built a **Pixhawk-RPI-based quadrotor** for crop health monitoring through Vision and NDVI based detection using RGB camera and NIR filters for local farms in Sindh, Pakistan.
- Developed a **mobile application interface** for real-time monitoring, command, and control of the quadrotor.

Phantom X Pincher PickPlace Package

- Designed and implemented a **custom perception-based pick-and-place pipeline** for the Phantom X 4DOF robotic arm.
- Developed the entire pipeline, without relying on external kinematics, motion planning, or control libraries/frameworks.

ALIF: Pretrained Urdu Generative Models

- Curated a **novel dataset of 6 billion tokens** for pretraining urdu language models.
- Developed **BPE based Urdu monolingual and Urdu-English bilingual tokenizers**, for efficient tokenization.
- Developed **instruction tuning datasets** for urdu language model.
- Developed, trained, and instruction-tuned decoder-based language models (100M and 1B parameters) for the Urdu

language, outperforming multilingual models of comparable sizes.

ENVISION for Visually Impaired Individuals

- Developed an accessible **mobile digital banking application** for Faysal Bank, specifically designed for **visually impaired users**.
- Secured **First Prize** at the Faysal Bank FinTech Innovation Hackathon 2024, winning a cash prize of PKR 500,000.

Extra-Curricular

Set Alumni Network (SAN) — Core Team Member

January 2023 - October 2025

The SET School

- Managed logistics and hosted the annual SET alumni reunion event 2023.
- Conducted a series of sessions at The SET School for high school students on trends and career pathways in AI.

Volunteer Developer

May 2024 - December 2024

The Bridge School

- Redesigned the website for the Bridge School with an enhanced UI, functionality and outlook.
- Introduced a cleaner database schema and developed a content management workflow for the website, improving site scalability and ease of updates.

Vice President — Maths Club

Habib University

May 2023 - May 2024

- Led a team of 8+ members to plan and conduct club activities including workshops, sessions and math talks.
- Co-led the core team to conduct Mathema'2023, Habib University's flagship Maths Olympiad for High School students.

Technology Instructor — Serve Club

Habib University

August 2022 - Dec 2022

- Designed a series of digital literacy workshops for the custodial staff of Habib University
- Taught basic elements of web designing and programming in HTML, CSS, Python.

Technologies

Languages: C++, C, Python, Matlab

Softwares/Frameworks: Simulink, CUDA, Creo, PyTorch, OpenCV, ROS2, TensorRT, Qualcomm AI Engine Direct SDK

Microcontrollers: Arduino, Raspberry PI, Nvidia Jetson/AGX, Qualcomm SoCs